Week 2 Lesson 1a: Introduction to Computer Ethics

Definition of Computer Ethics

Computer Ethics is the study of ethical issues that arise from the use of computers and information technologies. It examines the moral obligations of users and developers, and how computer-related decisions impact individuals and society.

James H. Moor (1985):

"Computer ethics is the analysis of the nature and social impact of computer technology and the corresponding formulation and justification of policies for the ethical use of such technology."

Why Computer Ethics Matters (Especially for Software Engineers)

- Software systems control healthcare, finance, transport, and even national security.
- Programmers often have access to sensitive data and may unintentionally or intentionally cause harm.
- Unethical computing can lead to:
- Privacy violations
- Systemic bias
- Identity theft
- Financial fraud
- Software failure causing physical harm

Example: A software engineer working on a smart health device must ensure the data collected remains private and protected from hackers.

Foundations of Computer Ethics

1. **Respect for Privacy**
    - Do not collect or share user data without consent.
    - Ensure data is stored securely.
2. **Honesty and Integrity**
    - Do not misrepresent code ownership or plagiarize.
    - Do not introduce backdoors or malicious code.
3. **Responsibility**
    - Be accountable for the software you write.
    - Report flaws and security risks even if it is costly.

4. **Fairness and Non-Discrimination**

- Avoid introducing algorithmic bias.
- Ensure software doesn’t discriminate based on race, gender, etc.

Core Ethical Issues in Computing

Ethical Issue	Description	Example
Privacy	Protecting users' personal data	Facebook–Cambridge Analytica scandal
Intellectual Property	Respecting copyrights, licenses, and patents	Using open-source code without attribution
Security	Ensuring software doesn't expose users to risks	Using outdated encryption methods
Digital Divide	Fair access to computing resources	Making systems usable only in English
Automation & Employment	Replacing human jobs with AI	Self-checkout systems replacing cashiers

Computer Ethics and Legal Frameworks

Ethical computing is supported by laws and policies:

Law	Purpose
Bangladesh ICT Act 2006	Regulates cybercrime, data misuse, and electronic transactions
Digital Security Act 2018 (BD)	Prevents illegal access, online defamation, and breaches of digital systems
GDPR (EU)	Ensures users’ right to consent and privacy
DMCA (USA)	Protects digital intellectual property

Key Legal Concepts:

- Unauthorized Access → Hacking, data breach
- Misuse of Data → Selling user data without permission
- Cybercrime → Spreading viruses, fraud via e-transactions

Ethical Theories Applied to Computing

1. **Consequentialism (Utilitarianism)**

Do what's best for the most people

Example: Implementing a patch immediately even if it costs the company, because it protects thousands of users.

2. **Deontology (Duty Ethics)**

Follow rules and principles regardless of the outcome

Example: Do not access user data without permission even if it benefits development.

3. Virtue Ethics

Focus on moral character and intention

Example: A virtuous developer practices transparency and honesty, even under pressure.

Real-World Scenarios for Software Engineers

Scenario 1: Data Breach in a Health App

Issue: Engineer discovers user health records are unencrypted.
Ethical Decision: Report it immediately, fix the vulnerability, inform users if necessary.

Scenario 2: Biased AI Algorithm

Issue: Facial recognition fails more on darker skin tones.
Ethical Decision: Retrain model using diverse datasets; flag the bias to stakeholders.

Scenario 3: Copying Code from GitHub

Issue: Developer copies a solution from a GitHub repo marked "All rights reserved."
Ethical Decision: Respect the license. Use open-source alternatives with proper attribution.

Role of Codes of Ethics

ACM Code of Ethics (Link: https://ethics.acm.org/code-of-ethics/)

- Contribute to society and human well-being
- Avoid harm
- Be honest and trustworthy
- Respect privacy
- Honor confidentiality

IEEE Code of Ethics (Link: https://www.ieee.org/about/corporate/governance/p7-8.html)

- Accept responsibility for decisions
- Avoid conflicts of interest
- Improve understanding of technology's impact